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Roll No

ME-505 (GS)
B.E. V Semester Examination, June 2020
Grading System (GS)
Dynamics of Machines
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. What do you mean by dynamical equivalent system? Explain.
2. a) Deduce an equation of maximum power transmitted by a belt drive system.
b) Explain the types of brakes.
3. A single cylinder double acting steam engine develops 150kW at a mean speed of 80rpm. The coefficient of fluctuation of energy is 0.1 and fluctuation of speed is $\pm 2\%$ of mean speed.
4. a) Derive an expression for the angular acceleration of the connecting rod of a reciprocating engine.
b) Define the terms coefficient of fluctuation of energy and coefficient of fluctuation of speed.
5. Differentiate between piston effort, crank effort and crank pin effort.
6. Discuss followers response analysis by phase plane method.
OR
For cams and followers discuss jump and cross over shock.
7. Each arm of a Porter governor is 300 mm long and is pivoted on the axis of rotation. Each ball has a mass of 6 kg and the sleeve weight 18 kg. The radius of rotation of the ball is 200 mm when the governor begins to lift and 250 mm when the speed is maximum. Determine the maximum and the minimum speed and the range of speed of the governor.
OR
Explain the uniform pressure theory and uniform wear theory.
8. Write short note on:
 - a) Types of followers
 - b) Dynamometer
 - c) Types of clutches
